

NICK AMBROSE

 San Diego, CA

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 602-350-8228

EDUCATION

Cal Poly, San Luis Obispo

B.S., Computer Science

GPA: 3.78

Magna Cum Laude

September 2020 - June 2024

SECURITY & CLEARANCE

**Active U.S. DoD Secret
Clearance**

**CompTIA Security+
Certified**

SKILLS & FRAMEWORKS

Flask

SQLAlchemy

Alembic

HTMX & Jinja

Docker & Docker Compose

AWS

**Authentication &
Authorization**

LANGUAGES

Python

SQL (PostgreSQL)

JavaScript

C

RISC-V Assembly

COMMUNITY SERVICE

Philanthropy Chair
Sigma Nu Fraternity

Communications Chair
Cal Poly Dance Marathon

WORK EXPERIENCE

Software Engineer | Moebius Solutions Inc. | September 2024 - Present

- Core developer on HARPs, a Flask-based document/workflow management web application for the U.S. Navy that automates creation, review, approval, and publication of Navy Training Systems Plans (NTSPs).
- Designed and implemented an Excel import/export framework mapping SQLAlchemy models to Excel templates and enabling schema-validated, idempotent bulk imports with conflict handling, reducing duplicate code and accelerating delivery of a highly requested feature.
- Led the migration of a legacy production database to PostgreSQL, moving 10M records in under 5 minutes while cleaning messy legacy data and converting free-form fields into normalized relational models.
- Implemented CAC-based certificate authentication for HARPs, validating client certificates, binding sessions to identity, and securing all request paths in a highly regulated government environment.
- Co-developed a reusable view/form framework that generates index/create/update/detail/delete views from SQLAlchemy models and Pydantic schemas, reducing boilerplate and improving consistency across HARPs in collaboration with a Flask core maintainer.
- Owned STIG compliance for HARPs, performing a full audit of Application Security and Development requirements and replacing spreadsheet-based tracking with scripted, maintainable Markdown documentation reusable across DoD projects.

Software Engineer | Better Texter Inc. | March 2023 - August 2023

- Built and shipped an AI-powered iOS app in Swift (UIKit/SwiftUI) integrating OpenAI chat models to rewrite and translate iMessage texts, reaching 200+ downloads.
- Implemented user feedback tracking with Google Cloud Firestore, capturing likes and dislikes on AI-generated rewrites to guide prompt tuning and product iteration.
- Partnered with a UI/UX designer to implement a streamlined onboarding flow and polished UI, improving usability and reducing friction for new users.

Computer Science Tutor | Cal Poly CS & SWE Dept. | Feb. 2022 - June 2024

- Mentored hundreds of CS students in algorithms, data structures, and object-oriented design, emphasizing clean, correct, and efficient code through hands-on debugging and problem solving.

PROJECTS

Automated Tee Time Booking System

- Built a serverless automation platform on AWS (Lambda, EventBridge, AWS SAM, Secrets Manager) to run time-critical booking workflows with secure session handling and fault-tolerant retries.
- Implemented a fan-out orchestration model with a controller Lambda coordinating course-specific workers that independently handle authentication and booking logic, improving throughput and isolating failures.

Threading Library

- Created a library that exposes an API for C programs to create and run threads.
- Implemented a round-robin scheduling algorithm to manage thread execution.